



SECP3213-01

BUSINESS INTELLIGENCE

PHASE 1

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Impact of Severe Weather Events on US Commercial Flight Operations:
A 2016 Big Data Analytics Approach

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1.0 Problem Background

The United States aviation industry scheduled more than 5.6 million flights in 2016, making it one of the most complex transportation networks in the world. Although information on schedules and weather forecasts is available, this information is normally segregated in different systems, making it impossible to have an integrated view on how atmospheric events impact operational disruptions. Weather-related delays remained a major obstacle to efficiency throughout that year, accounting for the bulk of non-technical delays recorded nationally.

1.1 Business Context

The magnitude of the challenge becomes evident when considering the volume of the data, with more than 1.1 million weather reports recorded from 2,071 weather stations across the country. It is a virtual impossibility for Airline Operational Managers and Airport Authority Planners to manually determine which regions of the country and which types of weather at specific regional airports create the largest national cascading effect without an appropriate tool. As reported by Airlines for America (2023), a single hour delay can cost an airline tens of thousands of dollars in wasted fuel, violations of crew time restrictions and passenger compensation. Additionally, without an enterprise-wide BI solution, these audiences are relegated to reactive decisions rather than proactive planning.

This project leverages 2016 flight performance data, high-granularity weather logs, and geographic airport metadata to transform raw operational figures into a strategic decision-support tool. This enables operational readiness through early identification of high-risk weather windows, targeted hub analysis across 341 primary commercial airports, and improved customer experience management through delay pattern awareness.

1.2 Importance of the Problem

The amount of weather disruption to the U.S. aviation system is large and measurable. Out of the 5,617,658 total flights recorded in 2016, approximately 50,143 flights experienced direct weather delays, with a combined total of 2,606,083 delay minutes accumulated across 2016. That averages out to about 52 minutes delay for every affected flight which is a significant cost of operations when considering that there are thousands of daily flights. In addition to the delays, there were 65,861 flights that were cancelled completely, many of which are caused by adverse weather conditions. Additionally, the lack of a formalized BI solution linking weather scenarios to flight performance data makes it difficult to determine which combinations of weather type, severity, and location most negatively affect the network efficiency. This creates a significant gap in predictive capability and operational decision support within the airline network management.

1.3 Expected Impact

By building a unified BI solution that links all 1,185,991 weather events to the corresponding 5,617,658 flight records through airport and date keys, this project is expected to deliver measurable decision-support value across multiple operational areas. First, identifying the airports and time periods that associated with 2,606,083 minutes of weather delay can help airline schedulers develop more effective buffers and minimize unnecessary delay propagation. Second, overlaying the 35,665 Heavy and 199,523 Severe events on a map of delay hotspots will assist airport operators in allocating infrastructure enhancements such as de-icing or storm response capabilities at the highest-risk airports. Third, being able to identify the delay trends associated with weather types such as Snow and Fog which have a combined total of more than 400,000 events, will allow airlines to enhance communication with passengers and develop proactive rebooking strategies. Overall, this project is an effort to prove that transforming raw aviation and weather data into a unified interactive BI dashboard can change decision making from reactive to predictive. Furthermore, it will also directly reduce the operational and financial consequences of weather delays throughout the U.S. aviation system.

2.0 Objectives

The first objective of this project is to combine the 2016 US flight performance data, weather event logs and airport geographical data layers into a single merged dataset in Alteryx Designer. Next, the dataset will be published to enable detailed analysis for Business Intelligence exploration.

The second objective is to identify the relationship between weather event types, severity levels, flight delay duration and cancellation rates at the top US airports in 2016. This analysis will help us to identify the weather conditions that create the highest operational risk for the national aviation system.

The third objective is to create an interactive BI dashboard in Power BI against temporal and geographic trends of weather-induced flight disruptions. The dashboard provides actionable insight and data driven implications to assist operational planning and strategic decision making for airlines and airport operators.

3.0 Dataset Description

3.1 Dataset 1: US Weather Events (2016–2022)

This is a structured dataset containing 8,627,181 records across 14 columns, spanning weather events recorded from January 2016 to December 2022. For this project purposes, only records from 2016 will be extracted, thus resulting in 1,185,991 usable records.

Link : <https://www.kaggle.com/datasets/sobhanmoosavi/us-weather-events>.

Column Name	Data Type	Description
EvenId	Object	Unique identifier for each weather event
Type	Object	Weather category (Rain, Snow, Fog, Storm, Hail, Cold, Precipitation)
Severity	Object	Intensity level of the weather event (Light, Moderate, Severe, Heavy)
StartTime(UTC)	Object	Date and time the weather event started
EndTime(UTC)	Object	Date and time the weather event ended
Precipitation(in)	Float64	Precipitation amount in inches
AirportCode	Object	Weather station identifier in ICAO format
LocationLat	Float64	Latitude coordinate of the weather station
LocationLng	Float64	Longitude coordinate of the weather station
City	Object	City where the weather event occurred
County	Object	County where the weather event occurred
State	Object	State where the weather event occurred
ZipCode	Float64	Zip code of the weather station location
TimeZone	Object	Time Zone of the weather station

3.2 Dataset 2: Airline Delay and Cancellation Data — 2016

This is a structured dataset containing 5,617,658 flight records across 28 columns, covering all domestic US flights operated throughout 2016.

Link :

<https://www.kaggle.com/datasets/yuanyuwendymu/airline-delay-and-cancellation-data-2009-2018>.

Column Name	Data Type	Description
FL_DATE	Object	Date of the flight
OP_CARRIER	Object	Operating airline carrier code
OP_CARRIER_FL_NUM	Int64	Flight number assigned by the carrier
ORIGIN	Object	Origin airport IATA code
DEST	Object	Destination airport IATA code

CRS_DEP_TIME	Int64	Scheduled departure time
DEP_TIME	Float64	Actual departure time
DEP_DELAY	Float64	Departure delay in minutes
TAXI_OUT	Float64	Taxi-out time in minutes
WHEELS_OFF	Float64	Time wheels left the ground
WHEELS_ON	Float64	Time wheels touched the ground
TAXI_IN	Float64	Taxi-in time in minutes
CRS_ARR_TIME	Int64	Scheduled arrival time
ARR_TIME	Float64	Actual arrival time
ARR_DELAY	Float64	Arrival delay in minutes
CANCELLED	Float64	Flight cancellation indicator (1 = cancelled)
CANCELLATION_CODE	Object	Reason for cancellation
DIVERTED	Float64	Diversion indicator (1 = diverted)
CRS_ELAPSED_TIME	Float64	Scheduled elapsed flight time in minutes
ACTUAL_ELAPSED_TIME	Float64	Actual elapsed flight time in minutes
AIR_TIME	Float64	Time in air in minutes
DISTANCE	Float64	Flight distance in miles
CARRIER_DELAY	Float64	Delay minutes attributed to the carrier
WEATHER_DELAY	Float64	Delay minutes attributed to weather
NAS_DELAY	Float64	Delay minutes attributed to national air system
SECURITY_DELAY	Float64	Delay minutes attributed to security
LATE_AIRCRAFT_DELAY	Float64	Delay minutes attributed to late arriving aircraft

3.3 Dataset 3: US Airport Metadata

This is a structured dataset containing 341 records across 7 columns, providing geographic and descriptive information for commercial airports across the United States.

Link : <https://www.kaggle.com/datasets/aravindram11/list-of-us-airports>.

Column Name	Data Type	Description
IATA	Object	Standard 3-letter airport code, used as the primary join key
AIRPORT	Object	Full name of the airport
CITY	Object	City where the airport is located
STATE	Object	State where the airport is located
COUNTRY	Object	Country where the airport is located
LATITUDE	Float64	Geographic latitude coordinate of the airport
LONGITUDE	Float64	Geographic longitude coordinate of the airport

4.0 Proposed BI Approach

4.1 How Alteryx and Visualization Tools Will Be Used

This project will make use of Alteryx Designer for data preparation and workflow automation, and Microsoft Power BI for the data visualization and dashboarding. Alteryx Designer will handle all data ingestion, cleaning, transformation and integration processes, producing a single enriched and analysis-ready dataset as the output. The result of this final dataset will be ingested into the visualization tool where interactive dashboards and storyboards will be built to communicate results to stakeholders. The integration of these two products delivers a true end-to-end BI pipeline from raw multi-source data to meaningful visual insights with no reliance on manual data handling.

4.2 Data Integration, Cleaning and Transformation Methods

The data integration process will involve connecting all three datasets through two sequential join operations within Alteryx Designer. The first join that will be performed is between 2016 flights and airports on IATA airport code to add airport name, city, state, latitude and longitude to each flight record. The result dataset will be then joined with the aggregate weather data on a composite key consisting of the IATA code and the flight date, to add the number of weather events, their severity and their type to each flight record. A left join will be used for the weather connection to ensure all flight records are retained regardless of whether a weather event was recorded on that day at that airport.

For data cleaning, the workflow will address several quality issues identified during initial exploration. The weather dataset uses ICAO airport codes with a leading "K" prefix (e.g., KATL instead of ATL), so a formula tool in Alteryx will be applied to strip this prefix and standardize all airport codes to the 3-letter IATA format. Missing values in delay-related columns such as WEATHER_DELAY, DEP_DELAY, and ARR_DELAY will be handled by

replacing nulls with zero where appropriate, as a null in these columns typically indicates no delay occurred. Duplicate weather event records will be identified and removed using the Unique tool in Alteryx to prevent double-counting during aggregation.

For data transformation, the weather dataset will be aggregated from event-level records to a daily airport-level summary using the Summarize tool in Alteryx. This aggregation will produce three derived fields with total weather events per airport per day, the most severe weather type recorded, and a concatenated list of all weather types observed, thus reducing the 1,185,991 weather records to a more manageable and join-friendly format. Lastly, flight dates and weather timestamps will both be parsed and standardized to a consistent date-only format to ensure accurate matching during the joint operation.

4.3 Planned Analysis and Visuals Forms

This analysis will aim to reveal the relationship between the weather and patterns of flight disruption over time, place and the severity of the weather. There are five visual forms identified as the most suitable forms of communicating these insights effectively.

A Bar Chart will be used to compare the average delay minutes in departures for the top US airports by weather severity level, and it will be the first visual form. This is an appropriate chart because it can be used to compare the performance of the airports side-by-side in various weather conditions, and can therefore be used to identify which airports are the most operationally vulnerable.

The second visual type is Line Chart where it will show the monthly trend of total minutes of weather delay and the number of weather events recorded in 2016. This type of form is suitable for showing seasonal trends and temporal relationships between weather activity and disruption to flights, for a 12-month period.

The third visual form is a Heatmap showing weather event frequency, by airport and month. This form serves as a useful tool for showing concentrations of weather activity in a concise and legible manner, as it will identify the locations and periods of time where the weather activity is greatest.

The fourth visual form is a Geographic Map, which will map delay hotspots throughout the United States based on the airport metadata dataset's latitude and longitude data. This spatial visualization is especially useful to planners at the airport authorities as it offers a first-hand geographic perspective of where disruptions are most focused throughout the national network.

The next visual form is a Scatter Plot which will explore the relationship between the number of weather events at an airport on a given day and the average number of minutes that the airport is delayed on the same day. This form is suitable for measuring and determining the direction of the weather-delay correlation and for determining the outlier airports with high weather activity and also high delays, or vice versa.